

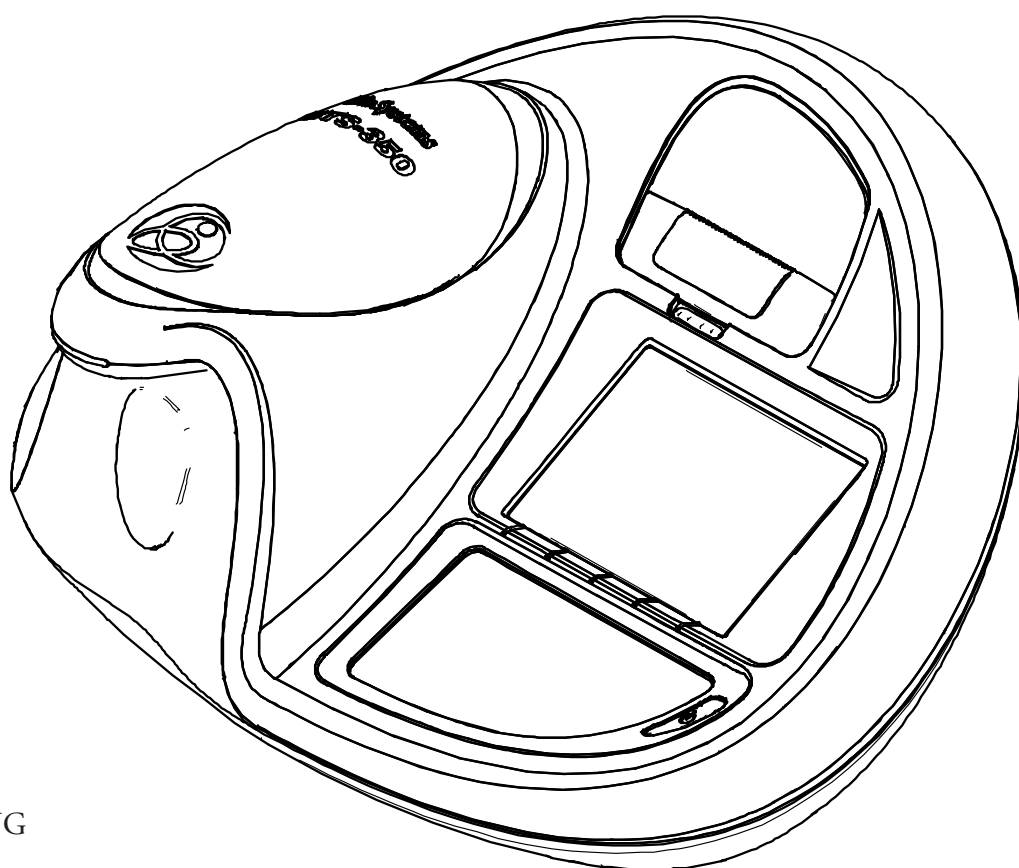
BTS-350

Y-350

Service

manual

English



TESE000010-03-ING

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The BTS-350 semi analyser is compliant with European Union directive 98/79/EC

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1. - Introduction

This semi-automatic analyzer BTS-350 is a laboratory instrument for performing biochemical and turbidimetric analyses for in vitro diagnostics.

This semi-automatic analyzer Y-350 is a laboratory instrument for performing chemical analysis specially for oenology.

Its compact design and the small number of components of which it is comprised allow for easy maintenance. The computer-based design has made it possible to eliminate mechanical adjustments in the optical system, simplifying installation and reducing maintenance. The use of easy-to-integrate, modern electronic components reduces the possibilities of failure and allows for the elimination of all electronic adjustments. A powerful programme allows for adjusting the different magnitudes handled by the equipment, using reference patterns or instruments and making corrections through software by numerical calculation.

At internal level, it has a new lighting system based on LEDs, guaranteeing the semi-automatic analyser a long life and very low maintenance.

The ergonomics of the housing have also been considered. The considerable height at which the Teflon tube emerges to suction the samples facilitates the use of the primary tubes.

The design of the power supply system is based on batteries, to preventing the loss of readings in the event of power cuts in the electric mains, and in the event of the laboratory not having a reliable electricity supply. This power supply system is an optional component.

The instrument electronics have been located in one plate, thereby allowing the printed circuit to be changed quickly and easily, in the event of a failure. Those circuits can be sent to the factory for repair.

This semi-automatic analyser is equipped with a large number of checking programmes to facilitate diagnosis and failure searches.

This manual has been designed not merely as a maintenance and repair guide for the equipment, but also as a document for training the Technical Assistance Service staff. The running principles as well as the electronic circuits are explained in order to obtain a global view of the instrument.

2. - Operating description

Below is a summary of how the semi-automatic analyzer functions to provide a clear picture of the unit and allow it to be studied in detail, in order to carry out maintenance and repair work.

These semi-automatic analyzers are based on the following functional modules:

- An optical system for performing photometric readings. (Patent pending).
- A suctioning system based on a peristaltic pump for positioning the sample in a continuous flow cuvette.
- A cuvette thermostating system to maintain all the reactions that require it at a constant temperature.
- A communications system that enables the semi-automatic analyzer to be connected to a computer which, if using a treatment programme, can use the data sent by the instrument.
- A microcontroller system that regulates the four modules described above.
- A battery supply system (optional).

A description of each of these functional modules is given below.

2.1. - Optical system

2.1.1. - Elements comprising it

The Figure 2.1 shows a diagram of the optical system, which is comprised of the following elements:

- A series of LEDs for each wavelength (1).
- A series of interferential filters (2).
- A series of light beamsplitters (3).
- Two lenses (4).
- One reference photodiode (5).
- One flow cuvette (6).
- One mirror (7).
- One main photodiode (8).

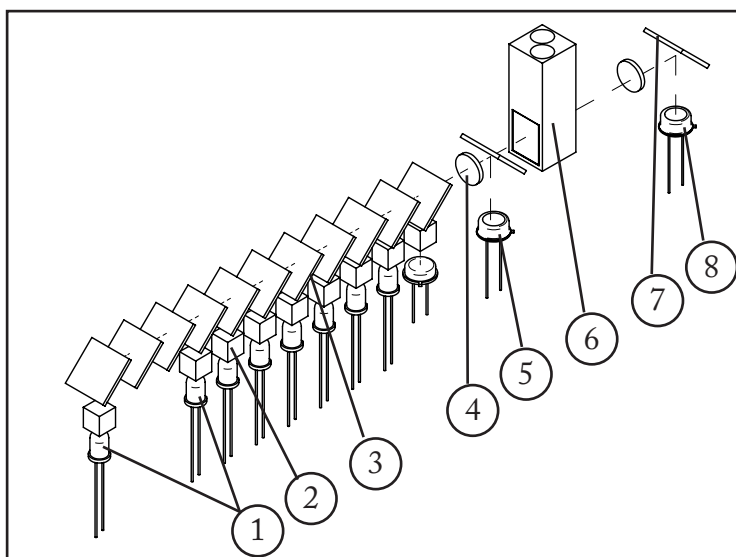


Figure 2.1

2.1.2. - Description of the system

The optical system is comprised of 8 LEDs, one for each wavelength (1). The microprocessor activates one LED, depending on the wavelength that is to be read. The microprocessor permanently controls that the LED is activated and also the current being supplied to it. The interferential filter (2) serves to define the spectrum width of the LED emission. The light beamsplitter (3) deflects the light beam towards the optical axis. This component is necessary because all the LEDs are placed perpendicular to the optical axis. Before crossing the sample, part of the light is deflected towards the reference photodiode (5) and the rest penetrates the sample through the flow cuvette (6). This photodiode measures the light before it penetrates the sample, thereby allowing potential fluctuations to be read. The light passing through the sample is deflected by a mirror (7) and then collected by the main photodiode (8), which makes the reading to calculate the absorbance.

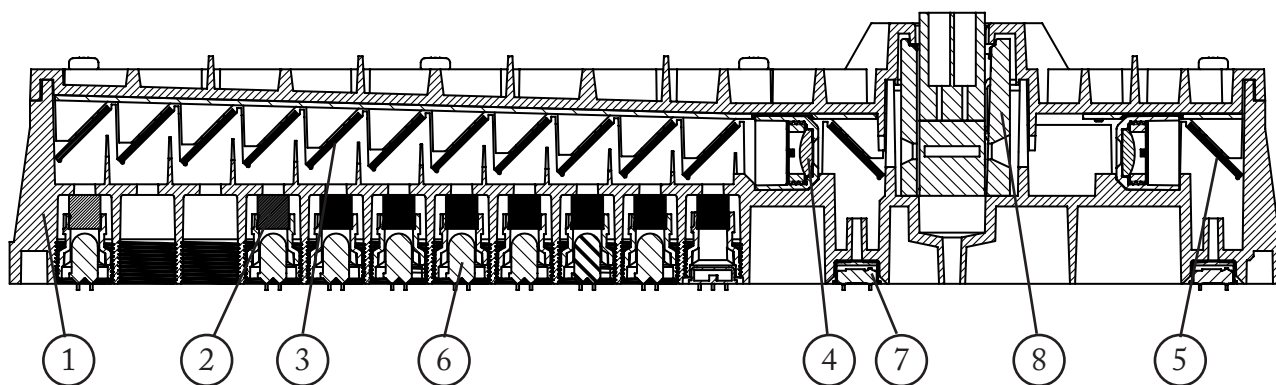


Figure 2.2

2.1.3. - Physical description

The optical system (see Figure 2.2) is comprised of a plastic-injected support (1) upon which all the components are mounted: interferential filters (2), light beamsplitters (3), lenses (4) and mirror (5). The interferential filters are placed inside a series of capsules and these are screwed to the optical bench support. Electronic components of the optical system: the LEDs (6) and photodiodes (7) are welded directly to the electronic board. The whole optical bench is also mounted on the electronic board, with the positions of the LEDs and photodiodes coinciding with those of the optical bench. The optical bench support holds the cuvette holder (8), into which the flow cuvettes or macro cuvettes are inserted from outside. The cuvette holder is comprised of a metal part to facilitate the thermostatisation of the flow cuvette.

2.1.4. - Adjustment

The optical system and A/D converter unit is linear, meaning it is not necessary to make any adjustments for absorbances. It is only necessary to adjust the integration times for each wavelength and the currents entering each LED. This adjustment serves to optimise the quantity of light that reaches both photodiodes. The objective is to have enough light to measure up to 3 absorbances without saturating either of the photodiodes. This adjustment is controlled by the A/D converter integration time and the current passing through the LED. See section 3.2.2. - Photometric parameters for the purpose of making this adjustment.

2.2. - Suctioning system

2.2.1. - Elements comprising it

In Figure 2.3 is a diagram showing the suctioning system, which consists of the following elements:

- One aspiration tube (1).
- One continuous flow cuvette (2).
- One peristaltic pump (3).
- One waste bottle (4).

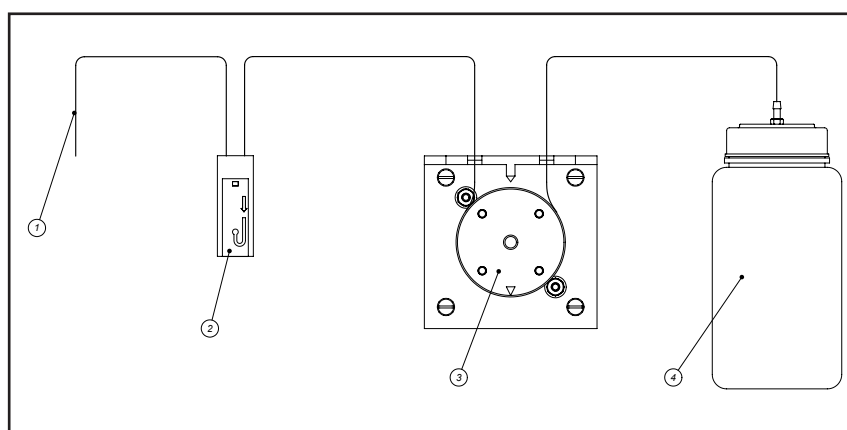


Figure 2.3

2.2.2. - Description of the system

The sample is suctioned through the aspiration tube (1). This tube, made of Teflon, has a predefined length for which the instrument is adjusted. The suctioned sample passes to the cuvette (2) where the readings are taken. The aspiration process is executed with a peristaltic pump (3) formed by a silicone-dispensing tube and a rotor with four rollers, operated by a stepper motor with a resolution of 200 steps per revolution. The sample finally passes through the pump towards the waste bottle (4) where it is discarded.

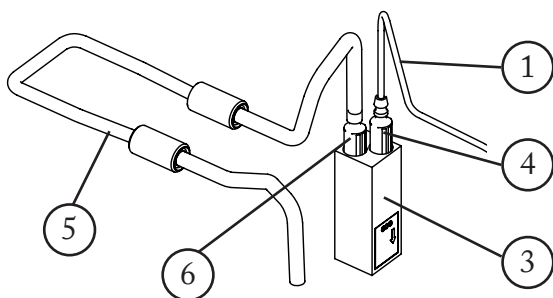


Figure 2.4

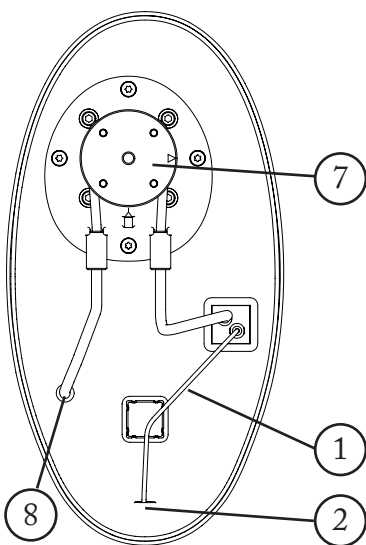


Figure 2.5

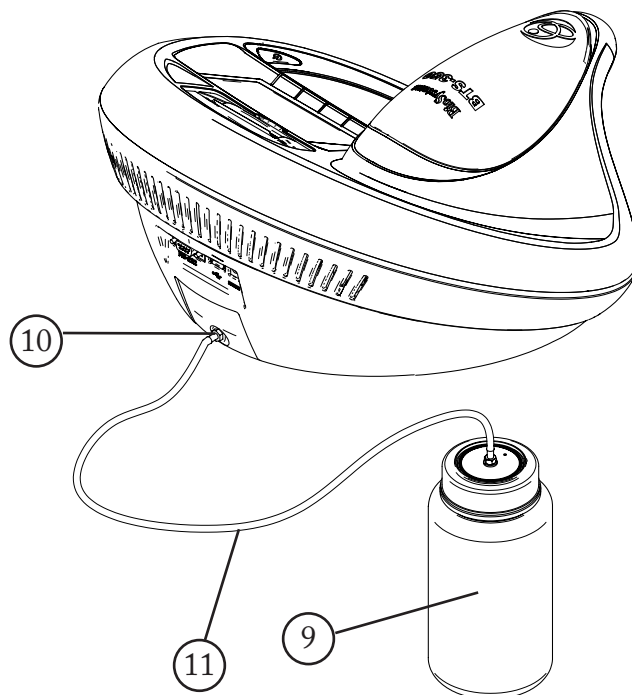


Figure 2.6

2.2.3. - Physical description

The suctioning tube (figures 2.6 and 2.7) is placed on the cuvette-holder tray. The suctioning tube (1) penetrates the housing through a rubber part (2) that serves as a guide. This tube is fixed to the cuvette (3) by an input fitting (4). On leaving the cuvette, it is connected to the dosing tube (5) through the output fitting (6). The dosing tube is housed in the peristaltic pump (7) and is finally connected to the waste output fitting (8).

The waste bottle (9) (Figure 2.6) is connected to the fitting (10) at the rear part of the appliance by the silicone tube (11).

2.2.4. - Control parameters

Three parameters control the suctioning function, and they must be programmed to obtain the expected performance characteristics. To adjust it, see section 3.2.3. - Aspiration system adjustment

- **Sample volume.** This is the number of pump steps that allows the volume of the sample to be suctioned to be adjusted.
- **Pump retard.** The number of seconds the pump will wait from the time the suctioning process finishes to the time when it is activated again to position the sample.
- **Positioning.** This is also the number of pump steps that determines the conveyance of the sample to the cuvette, ensuring that it remains in the correct position to be read.

2.2.5. - Programming

Programming the system is done by indicating the quantity of microlytes that must be suctioned. This will vary in each particular case. The appliance has several points where this value can be programmed. For instance, each test stored contains a volume of the sample to be suctioned.

2.2.6. - Adjustment

The nominal flow of the pump is 110 μL per revolution. However, the accuracy of this parameter depends on the tolerance of the length and diameter of the aspiration tube, with a deviation in the nominal value that varies depending on the degree of aging, for which reason the pump must be adjusted from time to time. The adjustment process is explained in section 3.2.3. - Aspiration system adjustment

2.3. - Thermostatisation system

2.3.1. - Elements comprising it

Figure 2.7 is a diagram of the thermostatisation system that consists of the following elements:

- One cuvette holder (1).
- One temperature sensor (2).
- One temperature sensor amplifier (3).
- The A/D converter in the microcontroller (4) and the part of the programme that controls the thermostatisation process.
- The Peltier cell power control circuit (5).
- One Peltier cell (6).
- One dissipator (7).
- One fan (8).

2.3.2. - Description of the system

The cuvette with the reaction mixture to be thermostatted is placed in the cuvette holder (1) with which it is thermally in contact. The cuvette holder is inserted in the optical support and in contact with one of the side of the Peltier cell (6). The other side of the cell is in contact with the dissipator (7). The Peltier pumps heat from one side to the other, depending on the direction of the current. The power control circuit (5) is in charge of making that current circulate in the adequate sense inside the Cell, in order to heat or cool depending on the microcontroller instructions. In the heating process, heat is pumped from the atmosphere (taken from the dissipator) to the cuvette holder and on cooling the opposite occurs. The radiator has a fan to assist in evacuating the heat. The temperature sensor (2) measures the cuvette holder temperature and, through the amplifier, (3) this is read by the microcontroller (4). This contains the thermostatisation programme and depending on the programmed temperature and value read, it activates the power control (5) heating or cooling as required.

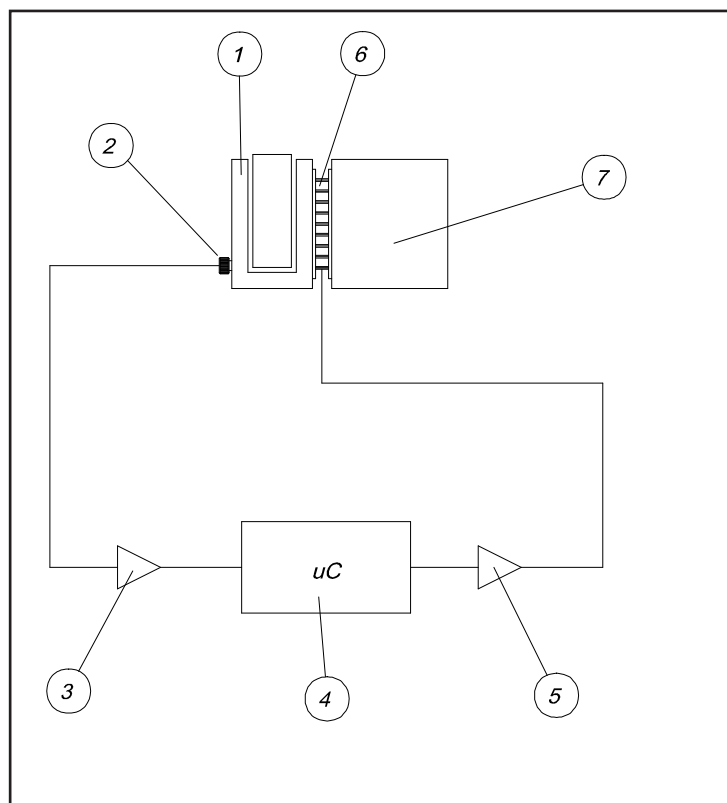


Figure 2.7

2.3.3. - Physical description

In Figure 2.8 a cross-section of the optical bench is shown, with the thermostatisation system. The cuvette holder (2) is inserted in the optical support (1). One of the sides of the Peltier cell (4) is touching the cuvette holder (2). On the other side is the radiator (5) and the fan (6). The temperature sensor (3) is housed in the cuvette holder.

2.3.4. - Programming

Programming the thermostatisation system consists of telling the instrument which temperature is desired for the reaction mixture. This will depend on each particular case. The appliance has several points where this value can be programmed. For instance, each stored test can programme a temperature.

2.3.5. - Adjustment

Like the other systems, the set of tolerances of the elements comprising the thermostatisation system includes a deviation between the programmed temperature and the real temperature, and so an adjustment process will be necessary to offset those deviations. The thermostatisation adjustment procedure is described in section.

2.3.6. - Precautions and maintenance

The thermostatisation system has two especially critical and delicate points: good thermal contact between both sides of the Peltier cell and the cuvette holder and radiator, respectively, and the thermal contact between the temperature sensor and the cuvette holder.

To guarantee proper functioning at these points, the following precautions should be taken:

- Place a thin but even layer of silicone on each side of the Peltier cell, covering the whole surface.
- The silicone must not exceed the edge of the cell surface or be between the two sides of it, as this would cause a thermal crossover, reducing the system's performance.
- The temperature sensor must have a layer of silicone that is sufficient to established good thermal contact with the bottom of the housing.

To assemble the above elements, following the procedures indicated on the CE15121 Peltier cell spares and AC15122 temperature sensor.

2.4. - Communications system

The semi-automatic analyzer are equipped with two communications systems:

- USB port type A
- USB port type B

2.4.1. - USB port type B

This is a USB port with type B connector.

The channel is supplied already preconfigured in the factory and has the following characteristics:

- Transmission speed (Baud rate): 9600
- Timeout: 0
- Terminal number: 0

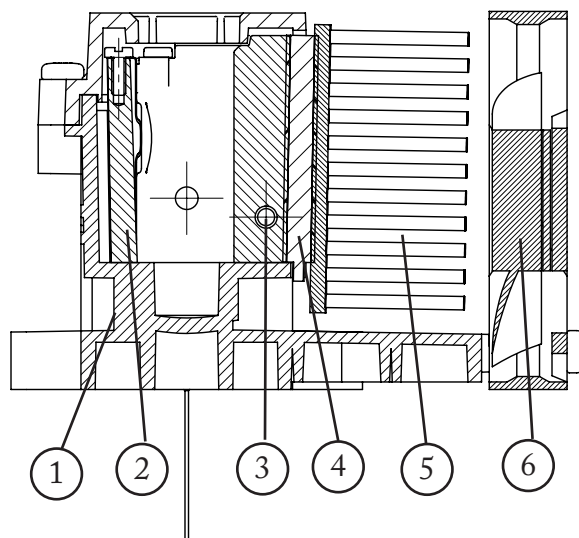


Figure 2.8

2.4.2. - *Communication characteristics*

To establish communication, it is necessary to have the PC-Photometer programme that is supplied together with the semi-automatic analyzer. The communications programme uses a send and wait protocol. Every time the transmitter sends a section, it waits to receive another from the receiver, confirming that it has received it correctly. If the transmitter fails to receive this section, it resends it after a certain time (timeout). If the receiver receives the section in error, it does not send the confirmation section, and the transmitter will resend the same section. The protocol uses an error-detector code, CRC-16.

This protocol functions as a MASTER-SLAVE system, and so it can communicate with several semi-automatic analyzers at one time. To do this, the semi-automatic analyzer must be identified by means of a terminal number. This parameter is configured with the communications configuration values. (MAIN MENU / CONFIGURATION / COMMUNICATIONS).

The communication is always started by the MASTER computer. The semi-automatic analyzer always remains alert and only responds if it receives a SLAVE information (REQUEST).

The information sent by the semi-automatic analyzer is the following:

- Quality control values
- Samples concentration values
- Updating the semi-automatic analyzer programme

To establish communication between the computer and semi-automatic analyzer, the appropriate link programme is required, which is executed from the computer. The programme stores the information received from the semi-automatic analyzer in a file that can be viewed from a text editor or using a spread sheet such as EXCEL.

2.4.3. - *USB Connector type A*

This is used to connect an external USB memory. To configure it, go to the CONFIGURATION/PRINTER menu and select the option USB. Every time the semi-automatic analyzer prints a document, it generates a results file. This is stored in the USB memory, instead of leaving through the printer. When printing a graph, the programme generates another file with the graph in bitmap format. The names of the files generated for the text file start with the letters PR, followed by the date in the YY-MM-DD format. Example: "PR090620.txt". A new file is generated every day. The name of the graph file has the following format: "CON00001.bmp". When printing out several graphs, the number of the file will gradually be increased.

2.5. - **Block diagram**

The block diagram gives a functional and global picture of the different electronic circuit components.

In Figure 2.9 there is a block diagram for that circuit.

2.5.1. - *LED control*

Circuit controlling the current of these LEDs. It is comprised of the CI: U14, U15 and transistors Q7 to Q18. The D/A converter output of the microprocessor generates a voltage of between 0 and 2.5V. This voltage enters the operational U14 and generates a current of between 0 and 30mA depending on the LED that is to be turned on. U15 is a decoder that polarises one of the 11 transistors to select a LED. The current programmed by the current source circulates through that LED.

2.5.2. - *Analogical-digital converter*

The circuit that makes the A/D conversion of both photodiodes. U18 is a double-ramp converter, with a current input. It is informed of the integration time through the length of the CONV signal. Once the device has completed the conversion,

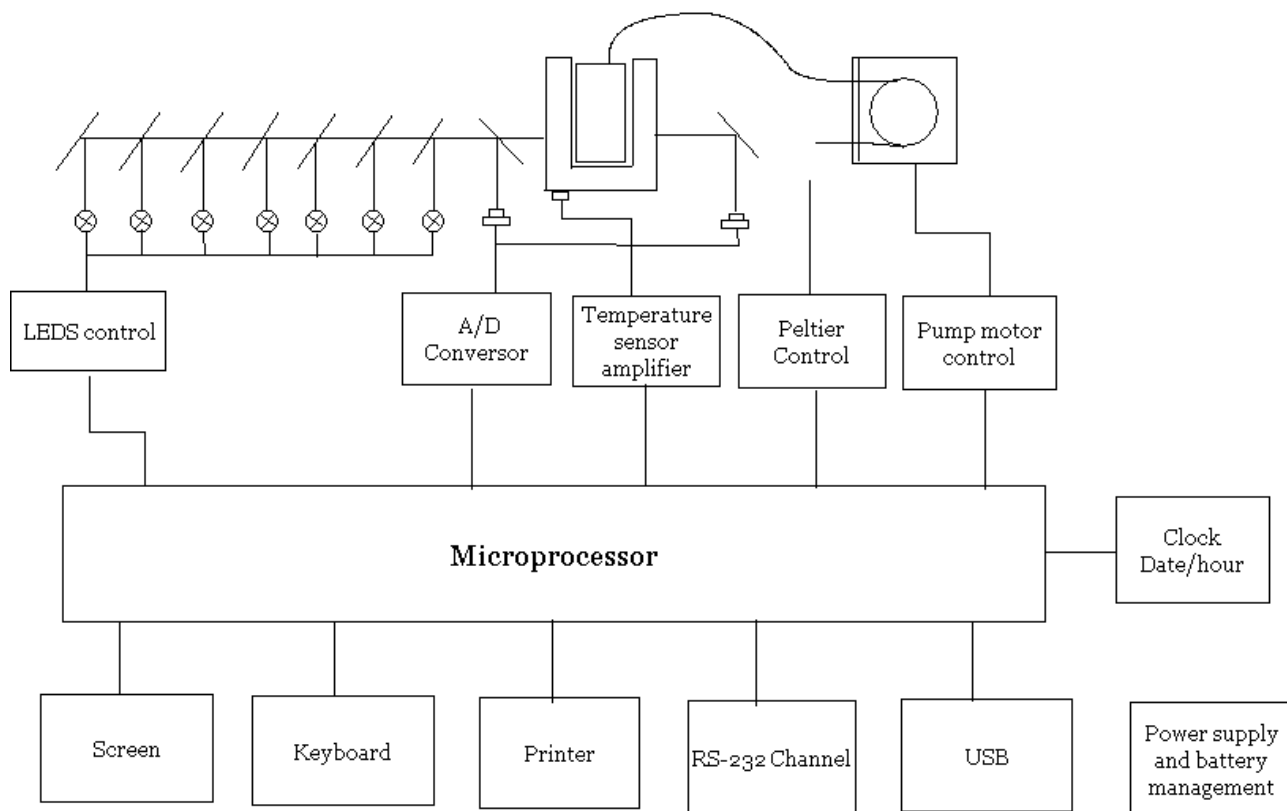


Figure 2.9

it indicates this through the DVALID signal. The microprocessor has the function of reading the account numbers through the DOUT and DCLOCK signals.

2.5.3. - Temperature sensor

The temperature sensor connects to J15. The sensor signal is converted to digital directly, through the microprocessor A/D converter.

2.5.4. - Peltier cell control

Circuits U1, U2 and U32, form a bridge of transistors in H. The Peltier cell is connected to the centre of that bridge. The bridge is controlled through the signals IN1 and IN2, and the control options are the following: Peltier cell deactivated, cell heating cuvette holder, Peltier cell cooling cuvette holder.

Connector J5 is connected to the fan that is mounted on the Peltier heatsink. When the Peltier cell has to cool the cuvette holder, this fan is activated through the *ActivaVentilador* signal. This fan also has another signal, *SensatVentPeltier*, that indicates the fan operation. If the fan breaks down, this is detected by the semi-automatic analyzer programme.

2.5.5. - Pump motor control

U21 is the circuit that controls the stepper motor to activate the peristaltic pump. The microprocessor generates the signals standard for circuit U21 to activate and deactivate the correct coil for making the motor turn, at any given time. Through the elements R97 and R98, the current in each coil is detected.

2.5.6. - Keyboard circuit

The keyboard circuit is comprised mainly of the keyboard itself, and a few protective diodes. The keyboard structure is a matrix of push buttons divided into rows and columns. Through the CPLD, the microprocessor generates the standard for activating the rows on the keyboard. The microprocessor reads the keyboard column signals directly, and decides which key has been pressed.

2.5.7. – *Printer control*

The printer is connected to the J10 connector. The power signals for the thermal points activated through the Q3 transistor arrive through this connector. The printer motor is activated through circuit U10. The data to be printed reach the printer through the signals *DataInPrinter* and *CLKprint*. The microprocessor reads the printer signals: paper sensor and printer head temperature sensor.

2.5.8. – *Screen circuit*

The CPLD, circuit U20, generates the standards for refreshing the screen. The display itself generates the V_{ee} voltage and that voltage is reduced to make it adjustable through transistors Q2 and Q23. This way, the display contrast can be controlled. The display backlight cable is connected to connector J13. It is supplied directly from the 5V.

2.5.9. – *USB connector type B*

The USB connector is controlled by circuit U34, which is a microcontroller specialising in managing the USB protocol. It is connected to the main microprocessor through a serial channel.

2.5.10. – *Fan control*

Both fans have an operation detection signal. The sensor in each fan is connected directly to the microprocessor.

2.5.11. – *Microprocessor*

This is responsible for linking and controlling all the semi-automatic analyzer systems, through its programme. The microprocessor system also contains the RAM memory, FLASH memory and control over an EPROM to load the programme in FLASH. It also contains the RESET generator circuit. It is comprised of circuits U11, U1, U2, F1, F2 and U8.

2.5.12. – *Day/time generator circuit*

The U13 circuit is the TRC. This circuit is responsible for generating and maintaining the semi-automatic analyzer date and time. This circuit needs a crystal to generate the internal frequency and a cell battery to maintain the date and time. It is connected to the microprocessor through bus I2C, signals SCL and SDA.

2.5.13. – *Temperature sensors on the plate*

The plate has 2 temperature sensors, U3 and U4. These are located near the power elements, such as the Peltier control drivers and power supply regulators. These sensors are connected to the microprocessor through the I2C bus.

2.5.14. – *Status indicator LEDs*

The semi-automatic analyzer has 2 status indicator lamps. One red and one green. Each lamp is activated through the Q19 and Q20 transistors and U33 decoder. The decoder is connected to the microprocessor through the I2C bus.

2.5.15. – *Buzzer and pump button*

The buzzer is activated through transistor Q1. The pump button, SW1, has a polarisation element R51 that is read directly by the microprocessor.

2.5.16. – *USB connector type A*

The USB connector is controlled by circuit U34, which is a microcontroller specialising in managing the USB protocol. It is connected to the main microprocessor through a serial channel.

2.5.17. – *Power supply*

This supplies the necessary voltages to the different parts of the appliance. There is an external power source that converts alternating current into continuous 15V current. The plate has a circuit, U23, that is a DC/DC converter, and converts the

continuous current at the input to 5V. The input voltage may come from the external power source or battery pack. There is a second DC/DC converter that converts the 5V voltage into 12V to supply the LED diodes and the peristaltic pump motor. The U35 circuit regulates the voltage to generate 3.3V voltage for the USB.

2.6. - List of board connectors and wiring diagram

Connector	Description
J1	USB type A connector
J2	Power supply connector
J3	Battery pack connector
J4	Main fan
J5	Peltier heatsink fan
J6	Display
J8	Peltier
J9	Peristaltic pump motor
J10	Printer
J11	USB type B connector
J13	Display backlight
J15	Temperature sensor
J17	Reserved
J18	Keyboard
J19	CPLD recording connector
JP1	FLASH/EPROM switch selection
SW1	Peristaltic pump push button
FAS1	GND connector
U1	EPROM socket

In the Figure 2.10 shows the wiring diagram of the board.

2.7. - Installing a programme release through EPROM

Steps to be followed for installing a new programme release:

- Open the top housing of the semi-automatic analyzer.
- Insert the EPROM with the new release in base U1 of the board. See the insertion of the EPROM into the base.
- Connect the two terminals to the JP1 connector on the plate.
- Turn on the semi-automatic analyzer.
- Wait for the recording process to end. This process takes a few minutes. A message will appear on screen indicating the process has ended.
- Turn off the semi-automatic analyzer.
- Carry out the same steps with the other EPROM memories.
- Remove the EPROM from the base and from the JP1 connector.
- Replace the top housing of the semi-automatic analyzer.
- Check the new programme release has been correctly installed. The initial screen shows the programme release.

2.8. - Battery pack (optional)

The battery pack is an optional element of the semi-automatic analyzer. This element is placed under the appliance in a compartment designed for this purpose, and connected to the main plate through the J3 connector.

That pack is comprised of a set of NiMH batteries and an electronic circuit to control the battery charge and discharge.

When the semi-automatic analyzer is plugged in to the main power supply and has a battery pack connected, the circuit charges the batteries to their maximum capacity.

If for any reason there is a failure in the main power supply, the semi-automatic analyzer electronics are automatically switched to the secondary battery power supply, without the user realising it.

The electronics of the charge circuit ensures that the batteries are not left to run down to the minimum, to prevent the battery cells from being damaged.

The battery capacity on full charge will supply power to the semi-automatic analyzer for two hours.

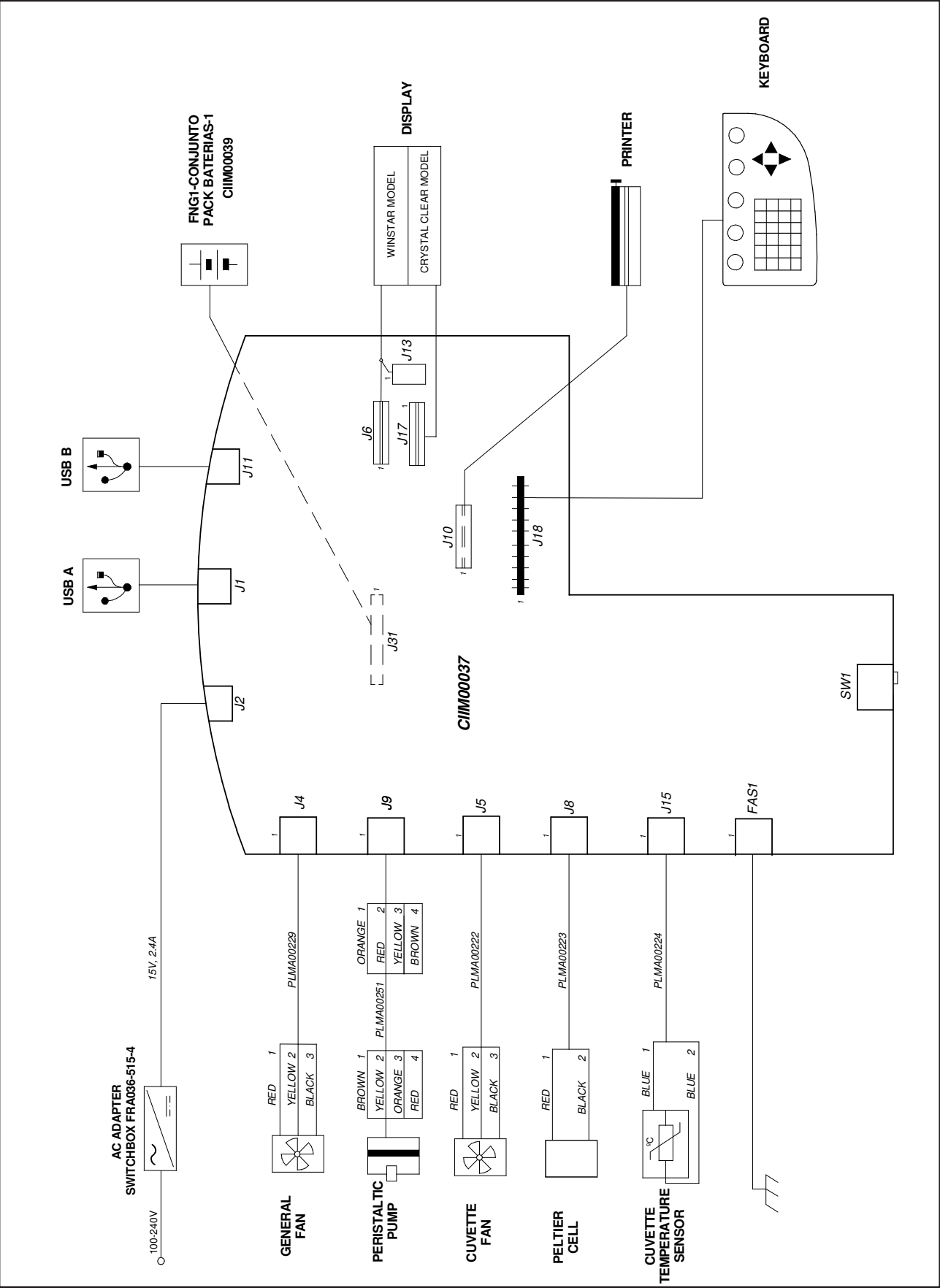


Figure 2.10

3. - Checking and adjustments

To facilitate the failures diagnosis and performing of adjustments to the instrument, this range of semi-automatic analyzers has a set of tests that make it possible to analyse the greater part of the appliance's functionality and assist in failure diagnosis. There are also programmes that permit the adjustment of the different systems.

Except for adjusting the peristaltic pump, which can be done by users, the tests and adjustments indicated can be accessed through the UTILITIES/SERVICE menu.

To enter those menus, the appliance requests an access code:

UTILITIES				29/03/09 17:03:19
PASWORD :				
				EXIT

Enter the access code (appendix V). A series of asterisks will appear on screen as you type. If you make a mistake type "C" to repeat.

3.1. - Service menu

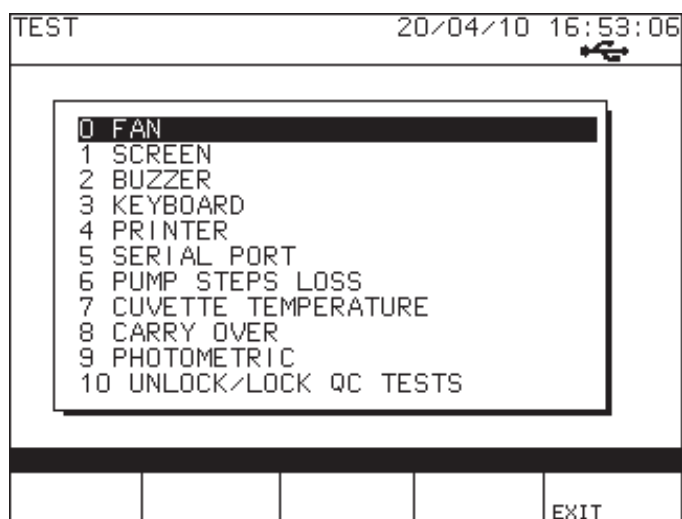
On selecting this option, the programme shows a screen like the one below:

SERVICE				29/04/09 12:30:42
0 TESTS 1 ADJUSTMENTS 2 UNLOCK/LOCK TESTS 3 UNLOCK/LOCK INSTRUMENT				
				EXIT

3.1.1. - *TESTS option*

Access the tests menu for failure diagnostics. These tests are listed below. Chapter 4 describes how they function.

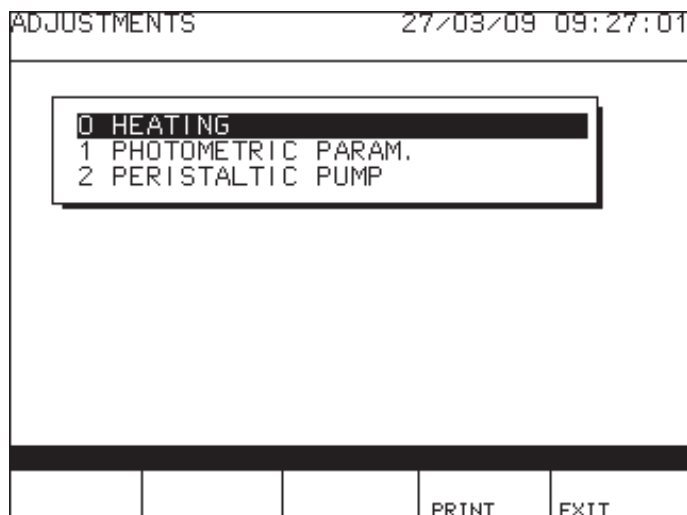
- FAN
- SCREEN
- BUZZER
- KEYBOARD
- PRINTER
- SERIAL PORT
- LOSS OF PERISTALTIC PUMP STEPS
- CUVETTE TEMPERATURE
- CONTAMINATION
- PHOTOMETRIC
 - LIGHT INTENSITY
 - DARKNESS
 - STABILITY OF READINGS
 - PRECISION
 - ACCURACY
- OPEN/CLOSE QUALITY TESTS



3.1.2. - *ADJUSTMENTS option*

The utility programmes are accessed to make the following adjustments:

- Thermostatisation
- Photometric parameters
- Peristaltic pump



3.1.3. - OPEN/CLOSE TESTS *option*

This option allows the user to open/close the tests, selecting them individually or as a whole.

In closed tests the parameters cannot be changed by the user.

3.1.4. - OPEN/CLOSE *appliance option*

This option opens/closes the instrument programming. On activating this option, the PROGRAMMING option in the main menu disappears and neither the tests nor the units can be changed. On opening the appliance again, the test change options are restored.

3.2. - Adjustments

3.2.1. - *Thermostatisation adjustment*

This procedure determines the method for adjusting the cuvette holder thermostatisation circuit. Both the cuvette holder thermostatisation circuit temperature sensor and the associated electronic circuit have a series of margins that must be offset to ensure that the cuvette holder temperature is adjusted to the programmed one. The adjustment makes it possible to calculate the necessary correction coefficients for performing the offsetting, through a calibrated thermometer.

3.2.1.1. - *Material required*

Calibrated digital thermometer with a resolution of 1 decimal, of the “Fluke 51” type, and temperature sensor.

3.2.1.2. - *General remarks*

N.B.: Bear in mind the corrections indicated in the digital thermometers when taking and recording temperatures.

3.2.1.3. - *Development*

Select the option using the path UTILITIES / SERVICE (Password) / ADJUSTMENTS / THERMOSTATATION.

ADJUSTMENT TEMP.		27/03/09 09:27:48		
NO ADJUSTED				
PELTIER ADJUSTMENT				
STEP	TEMPERATURE		ERRORS	
	READ	CURRENT	ABSOLUTE	RELATIVE
1	25	25	0	0%
2	37	37	0	0%
ADJUSTMENT FACTORS				
OFFSET		TANGENT		
0		0		
CHECK	ADJUST- MENTS	RESTORE ADJUSTM.	PRINT	EXIT

- Press the ADJUSTMENT function key. The message line will display “insert temperature sensor and press ENTER”.
- Insert the digital thermometer temperature sensor into the cuvette holder and press ENTER. The message line will read “Thermostating” followed by a time that is gradually reduced.
- On reaching zero, the following message appears: “Enter cuvette temperature at 25° C”. Enter the temperature value (corrected) that is marked by the thermometer and press ENTER. The following message will appear: “Thermostating” followed by a time that is gradually reduced.
- On reaching zero, the following message will appear: “Enter cuvette temperature at 37° C”. Enter the temperature value (corrected) marked by the thermometer and press ENTER.

3.2.1.4. - Explanation of the list concepts

- STEP: Each of the adjustment phases.
- TEMP. READ: Temperature read by the semi-automatic analyzer before applying the correction factors.
- REAL TEMP.: Temperature indicated by the thermometer that is input into the semi-automatic analyzer by the operator, through the keyboard.
- ABSOLUTE ERROR: Difference in degrees between the read and the real temperature.

$$E = T_{\text{read}} - T_{\text{real}}$$

- RELATIVE ERROR: Percentage of deviation in the real temperature compared to the read temperature.

$$E_{\text{relative}} = \frac{T_{\text{read}} - T_{\text{real}}}{T_{\text{real}}} \cdot 100$$

- OFFSET: Additive correction factor.
- TANGENT: Multiplicative correction factor.

3.2.1.5. - Check

This allows you to check the precision of the thermostatisation at a given temperature. To select this option, press the function key TEST In the UTILITIES / SERVICE menu (Password...) / ADJUSTMENTS / THERMOSTATATION. The cursor will go to the line:

TEMPERATURE TO BE TESTED: _

Insert the calibrated digital thermometer sensor in the cuvette holder, enter the temperature to be tested and press ENTER. The message line will read “Thermostatting” followed by a time that is gradually reduced. After waiting for 5 minutes (when the time counter reaches 0), the programme will ask for another temperature.

Now check that the temperature marked by the digital thermometer is within the acceptance criteria.

3.2.2. - Photometric parameters

This procedure serves to adjust the analogical-digital converter integration time values and those of the currents for each LED.

3.2.2.1. - Development

Select the option using the path UTILITIES / SERVICE (Password) / ADJUSTMENTS / PHOTOM. PARAM.

The following information will appear on screen:

ADJUST. PHOTOM. PARAM. 27/03/09 09:29:01					
LEDS	Int	DAC	DMX	PH1	PH2
340	172	91	100	1048575	1048575
560	350	115	125	1048575	845038
670	14	58	150	1048575	1048575
600	19	52	150	1048575	681368
535	60	100	150	1048575	616785
505	74	117	150	1048575	559501
635	35	100	150	777730	457796
405	440	137	150	1048575	1048575
---	120	90	255	4261	3999
---	120	90	255	4232	3988
850	16	124	150	1048575	890047
Press ENTER to read					
LED	Tint	DAC	MORE OPTIONS	EXIT	

- LEDS - indicate the wavelength of the emitting LED. Its position in the row indicates the order in which it is physically placed in the semi-automatic analyzer. As you can see the LED are not positioned in increasing order of wavelength. It is important to respect that order.
- Int - indicates Integration time (in number of counts). This is the time during which the analogical-digital converter is being integrated.
- DAC - indicates digital-analogical converter (in number of counts). It serves to establish the current value at the time of activating each LED.
- DMX - indicates Maximum value of the DAC (in number of counts). It serves to establish the maximum current to be used by the automatic system for adjusting the integration time and current of the LED. The ratio is the following: a value of 255 indicates a current of 50mA. This value is optimised in the factory to guarantee good emission and durability of the LED. It is advisable not to change this value.
- PH1 - indicates photodiode 1. Indicates the readings of the analogical-digital conversion of the main photodiode. It is indicated as the number of counts.
- PH2 - indicates photodiode 2. Indicates the readings of the analogical-digital conversion of the reference photodiode. It is indicated as the number of counts.

Function keys:

Press ENTER to read the selected LED. It shows the reading as the number of counts of the 2 photodiodes and the sample onscreen.

- LED - allows the label of a LED to be entered/changed. That label is the one shown during the programme when it is necessary to select a wavelength.
- Tint - Allows an integration time between 3 and 999 to be entered.
- DAC - Allows a DAC value to be entered for the current of the LED, between 0 and DAC MAX.
- DAC MAX - Allows the DAC MAX value to be entered, between 0 and 255.
- DELETE LED - eliminates the wavelength label. This wavelength will no longer appear when selecting the wavelengths in the rest of the programme.
- RESTORE CONFIGUR. - This restores the factory configuration for all the LED values, Tint, DAC and DMX.
- AUTOADJUST - Self-adjusts the Integration time and that of the DAC in the selected LED.
- AUTOADJUST ALL - This self-adjusts the Integration time and that of the DAC for all the LED.
- READ ALL - This reads all the LED.

3.2.3. - *Aspiration system adjustment*

This procedure indicates the steps to be taken to adjust the peristaltic pump flow and positioning of the sample in the flow cuvette. The pump flow depends on the number of steps of the motor and inner diameter of the dosing tube. This means that, with the same number of steps, different tubes have different flows, due to the tolerances of their inner diameters. Other factors that also influence the tiny variations in flow are the deformations that develop in the dosing tube with use, due to aging and the tolerances in length and diameter of the Teflon suction tube. This adjustment offsets the variations caused by the tolerances or aging of the tubes.

3.2.3.1. - *Required material*

- Tool for adjusting the volume
- 5 ml pipette (0.05 ml/division)
- Analytical precision balance (optional)

3.2.3.2. - *General remarks*

This procedure describes two alternative process methods: manual and automatic. It is advisable to use the automatic one. The function TEST allows a volume of fluid of between 100 and 5000 μL to be input, which are checked by aspiration cycles. On selecting this option, the cursor will go to the first line.

ADJUSTMENT PUMP		27/03/09 09:31:08		
NO ADJUSTED				
TEST VOLUME (μL):				
VALUE		CURRENT	NEW	
SAMPLE VOLUME		18000		
POSITIONING		800		
PUMP DELAY		2		
CHECK	MANUAL	AUTOMATIC	PRINT	EXIT

Type in the volume in μL that is to be checked and press ENTER. The message line will show "Insert tube with water and press PUMP".

Place a test tube with distilled water in the aspiration position and press PUMP.

3.2.3.3. – *Checking method*

- **Sample volume:** Fill a test tube with distilled water and weight it in an analytical precision balance. After the aspiration cycle, weight it again. The volume in μl is equal to the difference in weight in grams.
- **Positioning:** When the aspiration cycle is completed, check that the sample tail not entering the cuvette is approximately 5 mm (0-10 mm).

3.2.3.4. – *Manual mode*

Place the flow cuvette in the holder, ensuring that it is in the correct position.

- **SAMPLE VOLUME:** This corresponds to the aspired volume adjustment. Enter the number of quarter steps the pump must make to suction 5 ml and press ENTER. The theoretic value is 18340, equivalent to 1.09 ml per step (the motors operates in quarter steps: 0.2725 ml per quarter step). The number to be entered is an estimate and must be determined using the “trial and error” method through the function key TEST in the ADJUST PUMP menu.
- **POSITIONING:** This corresponds to the adjustment of the sample positioning. Input the number of quarters of step needed for the sample to be positioned into the cuvette with a tail of only 5 mm (0 – 10 mm) without entering into the cuvette and press ENTER. The theoretical value is 600 quarters of step. The number to be entered is an estimate and must be determined using the “trial and error” method through the function key TEST in the ADJUST PUMP menu.
- **PUMP RETARD:** The time elapsed between the suctioning cycle and the flow cuvette positioning cycle. Input the number of seconds (recommended 2) and press ENTER.

The manual adjustment is completed. To check it, use the volume adjustment tool, filled with water until the upper mark (3 mL).

- Press the function key CHECK.
- Input 2000 as sample volume (2 mL).
- Perform an aspiration cycle with the volume adjustment tool filled with water until the upper mark (3 mL).
- The sample tail that has not entered the cuvette is 0 – 10 mm long.
- The level of water remaining in the tool is between the two lower marks (equivalent to 1 mL $\pm$ one tolerance).
- Press EXIT: The programme will ask whether you wish to save the new values and shows a window for selecting YES or NO. Select one of these two options depending on whether you want to save the adjustment or not, and press ENTER. Press EXIT again to return to the main menu.

3.2.3.5. – *Automatic mode*

Place the flow cuvette in the holder, ensuring it is in the correct position.

- Press the function key AUTOMATIC.

Adjust sample volume:

- Press the function key SAMPLE VOLUME. The message line will show “Insert tube with 5 ml water and press PUMP”.
- Pipette exactly 5 ml of distilled water into a test tube and position at the sample input so that the end of the Teflon tube is at the bottom of the test tube and press the PUMP button.
- Approximately 4 mL are suctioned at normal speed and then it continues slowly. Then, the following message is displayed on the screen: “On suctioning the last drop press ENTER”.
- Carefully observe the bottom of the test tube and, when the last drop is drawn up into the suctioning tube, press ENTER.

Adjust position:

- Press the function key POSITION. The message line will show “Insert tube with water and press PUMP”.
- Position the test tube containing distilled water at the sample input and press PUMP. The pump is activated and, after a few seconds, the following message is displayed on the screen: “Remove the tube and press ENTER”.

- Follow these instructions. The pump is activated and the instrument calculates the position. Throughout this process the instrument performs photometric readings, so the cuvette carrier lid must be kept closed. The process takes about 70 seconds.

Checking the adjustment:

- Press the function key TEST.
- Input 2000 as sample volume (2 mL).
- Perform an aspiration cycle with the volume adjustment tool filled with water until the upper mark (3 mL).
- The sample tail that has not entered the cuvette is 0 – 10 mm long.
- The level of water remaining in the tool is between the two lower marks (equivalent to 1 mL $\pm$ one tolerance)
- Press EXIT. The programme asks whether you want to save the new values and shows a window for selecting YES or NO. Select one of these two options depending on whether you want to save the adjustment or not, and press ENTER. Press EXIT (F5) again to return to the main menu.

3.3. - Programming options

The semi-automatic analyzer has a series of functions that cannot be seen from the menu. These functions are activated through a combination of keys.

The process for accessing these functions is as follows:

- Turn off the semi-automatic analyzer.
- Press the key combination.
- Turn on the semi-automatic analyzer while those keys are pressed.

Key combinatoin	Function	Description
PUMP	Update of firmware change	Tool to access directly to the firmware update programme.
F1+7	Initialising of adjustments	This function allows you to initialise all the semi-automatic analyzer adjustment values. After doing this, the semi-automatic analyzer must be readjusted.
F2+8	Stressing	Function that performs 5-minute cycles. After each cycle the Peltier is activated, together with the peristaltic pump and printer. To exit that operating mode, press the Right cursor key.
F3+9	Change in serial number	Enters a screen that allows you to change the semi-automatic analyzer serial number.
Right cursor	Exit stressing	Exit stressing mode.

4. - Checking test

4.1. - Activation of a test

The program includes a set of test programs that enable the performance verification of diverse parts of the instrument and gives assistance in trouble-shooting. To perform a test, follow the path:

UTILITIES / SERVICE (Password) / TESTS

The tests available are described in the following sections.

4.2. - Fan

On executing this test, each time the ENTER key is pressed, successively run the following fans:

- The main fan
- The thermostatzation fan
- Stop both fans

4.3. - Screen

On executing this test, an operation is performed on screen every time ENTER is pressed. First, a series of horizontal strips are shown, then a series of vertical ones, then a table of characters and the contrast of the screen changes from the minimum to the maximum value. At the end of the test, the LED on the front of the appliance changes from red to green, in succession.

Check visually that the operations performed are uniform and that no anomaly is evident on the display.

4.4. - Beep

When performing this test, the internal beeper sounds several times.

4.5. - Keyboard

Each time a key is pressed, the beeper sounds and the key is displayed. To exit press the EXIT key (F5) twice.

4.6. - Printer

A set of characters is printed and then 10 rows of asterisks with the 10 programmable intensities.

Check that the characters listed are clear and legible and that the scale of intensities is growing and regular.

4.7. - Serial channel

Connect a USB cable to a computer. Execute a hiperterminal in the computer and configure with the appropriate COM.

Each time a key is pressed, the corresponding character is shown in the section of the display corresponding to the characters sent, and the same character should appear in the section of characters received. Exit by pressing F5.

Through this test it is possible to modify the communication configuration parameters.

4.8. - Loss of peristaltic pump steps

To perform the test, proceed as follows:

- Turn the roller holder with your hand until the arrow marked on it coincides with the one on the support.
- Press ENTER. The pump will make several turns.
- Check that the turn is performed on centre and with no abnormal noises.
- After stopping the roller holder, check that both arrows maintain the same position opposite each other.
- Press ENTER several times for the pump to perform successive operations.

4.9. - Cuvette temperature

To check the thermostatisation system operation, proceed as follows:

- Insert the temperature sensor of the digital thermometer into the cuvette holder.
- When requested by the programme to do so:

TEMP. CUUVETTE TEST		30/03/09 08:22:57	
TEMPERATURE TO CHECK: —			
			EXIT

- Insert the test temperature.
- After five minutes of thermostatisation, the programme will ask for a new test temperature.
- Check that the temperature indicated by the digital thermometer is within the acceptance criteria.

4.10. - Carry over

This enables carry over in the suctioning system produced by the consecutive reading of two different samples to be studied, in which the first usually has a high absorbance and the second, a low one.

Select the values of the following parameters to perform the test:

- Reading wavelenght.
- Stabilisation time (1).
- Sample volume (400).

To perform the test:

- Press F1 to begin the readings.
- Enter the baseline and then the first sample five times (L1 to L5) and the second, five times (L6 to L10).
- The semi-automatic analyzer calculates the carry over, based on the formula:

$$\text{Contamination} = \left(\frac{L6}{X_{\text{med}}} - 1 \right) \cdot 100$$

$$X_{\text{avg}} = \frac{L7 + L8 + L9 + L10}{4}$$

4.11. - Photometric

4.11.1. - *Light intensity*

This enables the user to know the current generated by the photodiode for a quantity of light received, thereby making it possible to know the light intensity of the instrument for each wavelength.

The semi-automatic analyzer reads the light intensity for each of the programmed wavelengths. This test can be done without cuvettes (light sensitivity to air) or with cuvettes. If using a flow cuvette, it must be filled with distilled water with a WASH cycle.

The light intensities read must be within the ranges indicated in the tables of Appendix I.1, if measuring the light intensity with the flow cuvette filled with water.

LIGHT INTENSITY		27/03/09 09:58:52	
LEDS	TINT	MAIN	REFERENCE
340	14	985921	134196
405	6	1048575	511752
505	13	953716	466502
535	49	964252	500846
560	149	961015	352907
600	69	960648	504005
635	45	971689	571516
670	220	984119	579956
850	11	974616	611641
			EXIT

4.11.2. - *Darkness*

This makes it possible to verify the darkness current of the reading system.

Check that the housing and lid are well closed to prevent the light from entering.

DARKNESS		27/03/09 10:00:13		
MAIN				
Tint	Nmax	Nmin	Aver.	Std. Dev.
6	4165	4082	4123	21
15	4172	4099	4136	21
50	4228	4130	4177	26
250	4428	4332	4387	26
500	4735	4163	4542	143
20	4183	4097	4130	20
Press ENTER				
MAIN	REFERENCE			EXIT

The instrument does not select a wavelength. Take 30 readings with different integration times and calculate the following parameters:

- Maximum number of counts
- Minimum number of counts
- average
- standard deviation

Take the readings for the main and reference photodiodes. Press F1 to see the results for the main photodiode and F2 for the results of the reference photodiode.

The test calculates using the following integration times, by default:

6, 15, 50, 250 and 500

On starting the test, the programme requests an integration time for which the user can select an integration time other than the pre-established ones for taking the reading.

See Appendix I.2 for the acceptable values range.

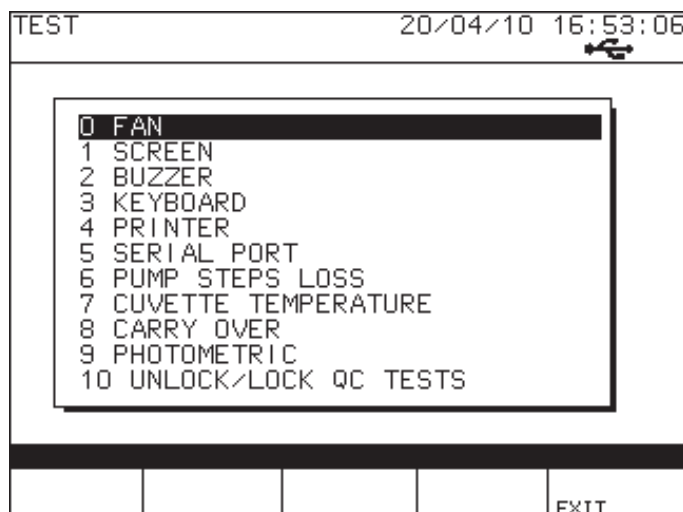
4.11.3. - Readings stability

This allows the stability of a set of repeated readings made on one sample to be analysed.

When beginning the test, the instrument requests:

- Reading wavelength (340).
- Stabilisation time (1).
- Sample volume (400).
- Number of intervals (30).
- Interval time (1).

The values between brackets are the ones recommended for a routine test.



Programme these parameters depending on the type of sample to be read. The interval time and number of intervals time will give the length of the test to be performed.

Press F1 to begin the test. The appliance asks to enter the baseline and then the sample to be read. If this is the same fluid used for the baseline, a zero stability test can be performed.

For a routine test, make the BASELINE with standard n° 0 of the AC10495 standards kit and use standard n° 4 as the SAMPLE.

The reading n°, time and absorbance are printed out. The test statistical data are printed at the end: maximum absorbance, minimum absorbance, average and coefficient of variation (CV).

See Appendix I.5 for the acceptable values range.

4.11.4. - Precision

This allows successive readings to be made, changing the sample each time. If using the same fluid, it allows the precision of the appliance to be tested (replicability between identical samples).

When beginning the test, the instrument requests:

- Reading wavelenght.
- Stabilisation time.
- Sample volume.
- Number of intervals.

Press F1 to begin the test.

The appliance asks for the baseline to be entered and then as many samples as there are numbers of intervals programmed. The test statistical data are printed at the end: maximum absorbance, minimum absorbance, average and coefficient of variation (CV).

4.11.5. - Trueness

This allows the accuracy of a sample with a known value to be checked.

WARNING: To guarantee the test is reliable, the known value of the sample must have been determined with a very exact reference method.

When beginning the test, the instrument requests:

- Reading wvaelenght.
- Stabilisation time.
- Sample volume.

- Temperature.

Press F1 to begin the test.

ACCURACY		27/03/09 10:04:30	
Reference: 19			
Absolute Error:		0.191	
Relative Error:		0.001	
		0.467%	
Input reference value			
		PRINT	EXIT

Enter the baseline Then the instrument asks for the reference value. Insert the sample theoretical value by means of the numeric keyboard. When requested by the programme to “Enter SAMPLE”, enter the known sample. The instrument prints the following data:

- Reading number.
- Absorbance.
- Absolute errors:
- Relative errors:

4.12. - Open / Close QC tests

On leaving the factory memory positions 150 to 155 contain the tests used to perform the final quality control of the finish. These tests are de-activated in the factory; if they are activated again, the names of this tests will appear again in the list of programmed tests.

This option is for internal factory use and is of no interest to the Technical Assistance Service.

5. - Maintenance

5.1. - General care

In order to ensure optimal operation of this instrument it is necessary to follow some minimal maintenance rules.

- Never use detergents or abrasive products for cleaning the outside of the instrument. Use only a cloth dampened in water and neutral soap.
- If a reagent or any corrosive product is spilt on the apparatus, clean it immediately with a damp cloth and water.
- The cuvette holder tray is equipped with a watertight joint in order to prevent penetration of liquid into the inner part of the instrument. If liquid is spilt onto the tray, wipe it with a damp cloth and water.
- If a cuvette breaks in the cuvette holder or liquid pours into it for any reason, the holder has a drainage hole connected to the outside. However, distilled water must be passed through it, and then the inside dried.

5.2. - Cleaning of the optical components

The following indications should be taken into account for the cleaning of the optical components:

- Optical components to be considered: lenses, filters and photodiodes.
- Recommended material:
 - WYPALL LITE paper (56 gr/m²) SCOTT.
 - 50% alcohol + ether solution.
 - Cotton buds.
 - Blower.
- To ensure correct handling of the optical components, please take the following precautions:
 - The area for manipulating the instrument should be clean and tidy.
 - As the components are fragile, they should be treated carefully; a fall could result in breakage.
 - Avoid touching the active parts with your fingers.
 - Lenses, filters and photodiode should be held by their sides, LEDs and photodiodes by the connecting terminals.
 - To clean the components, first remove the dust using the blower; thus scratches caused by small particles on the surface can be avoided, when rubbing with paper.
 - In the case of persistent or greasy dirt, clean with a paper soaked with the alcohol/ether solution and then with a dry paper. Sometimes, for instance when cleaning the filters or the photodiode window, cotton buds may be helpful together with the paper in the most delicate parts.
 - After any cleaning it is convenient to repeat blowing with the blower thus eliminating any residual paper or cotton nap.
 - When mounting or dismounting any component take into account the corresponding tools and procedures, as they are designed to avoid manipulation problems.

5.3. - Cleaning the filters

- Dismantle the top housing of the semi-automatic analyzer.
- Dismantle the optical bench of the electronic board.
- Remove the optical bench filters at the lower part of the bench.
- Clean them as indicated in section 5.2.
- Reassemble the filters in their holders and the latter on the optical bench.

5.4. - Cleaning the lenses

- Dismantle the top housing of the semi-automatic analyzer.
- Dismantle the optical bench cover. Be careful when unscrewing and screwing the optical bench cover, as the plastic material does not allow this operation to be performed several times. After performing it several times, the screw thread will become worn. **Warning** Be careful when dismantling the optical bench cover that the mirrors and beamsplitter are not caught on the cover when it is lifted.
- Remove the lens holder and lenses.
- Clean them as indicated in section 5.2.
- Reassemble the lenses in their holders and the latter on the optical bench. When reassembling the cover, be careful to keep all the mirrors and beamsplitters in their place. They must not be interchanged.

5.5. - Cleaning the photodiodes

- Dismantle the electronic board.
- Dismantle the optical bench of the electronic board.
- Clean the photodiodes as indicated in section 5.2.
- Reassemble the optical bench and electronic plate.

5.6. - Cleaning the aspiration system

- It is necessary to clean the SUCTIONING CIRCUIT properly after each series of measurements and at the end of the day.
- On finishing a series of measurements, wash the suctioning circuit with abundant distilled water.
- At the end of the working day, wash thoroughly with a detergent solution such as the one provided with the instrument. Lastly, rinse with distilled water and empty the circuit by performing wash cycles with air.
- Finally, to maximise the life of the peristaltic pump tube, it is advisable to take it out from its mounting, so that it remains loose and without tension. On starting a new work session, re-insert it in its place.
- If the outside end of the suction tube has deteriorated, a few millimetres may be cut making a perpendicular and clean cut. We recommend using a cutter, not scissors, to prevent the tube from being nipped. In this case, the POSITIONING parameter must be readjusted.
- Replace tubing by a new one in case of deterioration. Always use original spares.

5.7. - Cleaning the flow cuvette

Cleanliness of both the outside and the inside of the flow-cuvette is very important. Proceed as follows:

To clean the inside, proceed as described in section 5.6.

To clean the outside, use alcohol and then dry with a soft paper (section 5.2).

5.8. - General cleaning of the instrument

It is important to prevent dust from building up inside the instrument that could affect the optical system. Carefully remove all dust from the inside of the instrument.

5.9. - Dismantle the top housing

To remove the top housing, proceed as follows:

- Turn the semi-automatic analyzer.
- Unscrew the 8 torx screws.
- Separate the top housing carefully from the rest of the appliance.

5.10. - Dismantle the electronic board

To access the electronic board, proceed as follows:

- First dismantle the top housing.
- Dismantle the display, by unscrewing the 4 screws at its ends. Unplug the 2 connectors from the display.
- Unscrew the 4 screws of the printer support. Two screws are on the outer part of the support, and the other two are inside it. Unplug the connector from the printer.
- Unscrew the 3 screws on the tray. Unplug the motor connections from the peristaltic pump and from the motor earth clamp.
- Unscrew the screw of the light guide on the front panel.
- Unscrew the 4 screws that join the electronic board to the semi-automatic analyzer.
- Carefully remove the board from the appliance, together with the optical bench.

5.11. - Semiautomatic analyzer Firmware Update

5.11.1. *Updating the firmware from a USB memory*

Steps to install the new firmware:

- Connect the USB memory with the new version of the firmware stored in the main folder
- Select the menu: *Utilities / new installation program*
- You access the program monitor (internal program to manage change firmware)
- Select *F3*, Access to USB
- Select the first option, *firmware load file*
- A menu with all files that can be loaded appears. Select the file with the new version.
- The upgrade starts in the semi-automatic analyzer. This process takes a few minutes.
- When finished, on screen appears a message of *complete download*.
- Press the *exit* button (*F5*) twice
- The semiautomatic analyzer is initialized with the new version.
- Check on the home screen version of firmware.

Next table indicated the firmware type to use depending on the instrument model and the version of the electronic board.

Board number	Instrument model	File Type
CIIM00037-07	BTS-350	File with extension .bin
CIIM00037-08	BTS-350	File with extension .b35
CIIM00037-09	BTS-350	File with extension .b35
CIIM00037-08	Y-350	File with extension .y35
CIIM00037-09	Y-350	File with extension .y35

5.11.2. *Firmware update via EPROM memory*

Steps to install the new version of the program:

- Open the top cover of semiautomatic analyzer
- Put the EPROM with the new version in socket U1 of hte board. Note the orientation of the EPROM in the socket.
- Attach the two terminals of the connector JP1 of the board.
- Turn on the semiautomatic analyzer
- Wait until the recording process finish. This process takes a few minutes. On the display will appear a message indicating that the process it has completed
- Turn off the semiautomatic analyzer
- Do the same with the second EPROM
- Remove the EPROM socket and union connector JP1

- Place the upper housing semiautomatic analyzer
- Check that you have correctly installed the new version. On the Initial screen displays the software version.

5.11.3. *Firmware update via PC-photometer program*

Steps to install the new firmware:

- Connect the USB cable to the computer semi-automatic analyzer.
- Select the menu: *Utilities / new installation program*
- You access the program monitor (internal program to manage change firmware)
- Select the option *communications / Load new program* of PC-photometer programme
- Select a file to send.
- Press the button to *start*
- The upgrade starts in the semi-automatic analyzer. This process takes a few minutes.
- On the display will appear a message indicating that the process it has completed
- Press the *exit* button (F5) twice
- The semiautomatic analyzer is initialized with the new version.
- Check on the initial screen the version of firmware.

5.11.4. *Recovering a failed firmware update*

If an error has occurred during the firmware upgrade process or has not been completed successfully or a power cut has occurred, follow these steps:

- Turn off the semiautomatic analyzer.
- Turn it back on by pressing the PUMP button.
- This process will re-access the program part of the update process
- Ensure that the upload file is correct and not corrupt.
- Repeat the update process again.

5.12. - Spares parts

An explanation of how to change each spare in the semi-automatic analyzer can be found among the documents included with each spare. A copy of all the documents and the service manual can be found in the cd-rom.

Appendix I. - Adjustment value margins

I.1. - Integration times and DAC for each wavelength of the filters

These values are obtained with the flow cuvette filled with distilled water.

Integration times

BTS-350		
WAVELENGTH	MINIMUM VALUE	MAXIMUM VALUE
340 nm	6	94
405 nm	6	77
450 nm (optional)	6	84
505 nm	6	55
535 nm	6	154
560 nm	6	110
600 nm	6	110
635 nm	6	66
670 nm	6	176

Y-350		
WAVELENGTH	MINIMUM VALUE	MAXIMUM VALUE
280 nm	6	30
340 nm	6	30
405 nm	6	30
420 nm	6	110
505 nm	6	110
520 nm	6	80
560 nm	6	20
620 nm	6	40
635 nm	6	20
670 nm	6	50
750 nm	6	20

DAC Analogical/Digital Converter

BTS-350		
WAVELENGTH	MINIMUM VALUE	MAXIMUM VALUE
340 nm	50	90
405 nm	50	130
450 nm (optional)	75	140
505 nm	75	140
535 nm	10	140
560 nm	15	120
600 nm	75	140
635 nm	30	140
670 nm	20	140

Y-350		
WAVELENGTH	MINIMUM VALUE	MAXIMUM VALUE
280 nm	80	130
340 nm	60	130
405 nm	100	130
420 nm	100	130
505 nm	100	130
520 nm	100	130
560 nm	20	130
620 nm	60	130
635 nm	60	130
670 nm	40	130
750 nm	30	130

I.2. - Darkness

These margins serve for both the principal photodiode and for the reference photodiode.

Integration times	MIN. VALUE	MAX. VALUE
6	3500	4500
15	3500	4500
50	3500	4500
150	3500	4500
500	3500	4500

I.3. - Thermostatisation adjustment

After calibrating the thermostatisation, the temperature must be maintained with a stability of:

$$\pm 0.2^{\circ}\text{C}$$

I.4. - Adjusting the peristaltic pump

The sample tail that must be inside the suctioning tube without entering the flow cuvette is:

$$0 \text{ to } 15 \text{ mm}$$

If these values are not obtained, change to the MANUAL procedure.

I.5. - Acceptance range in the stability test

Level	CV
Neutral Filter N° 1 (~0.2 ABS) a 505nm	<1 %
Neutral Filter N° 4 (~2 ABS) a 505nm	<0.1 %

Appendix II. - Default configuration

CONFIGURATION

LANGUAGE	: ENGLISH
SCREEN CONTRAST	: 5
DEFAULT PRINTER	: ACTIVATED
PRINTER CONTRAST	: 7
REPORT HEADLINE	: SUMMARISED
BUZZER	: ACTIVATED
WASHING MODE	: 0
PATIENT CODE	: DEACTIVATED
LABORATORY NAME	: (blank)

WAVELENGTH TABLE

1.....	340
2.....	405
3.....	505
4.....	535
5.....	560
6.....	600
6.....	635
7.....	670

UNITS TABLE

0 -	mg/dL
1 -	U/L
2 -	g/L
3 -	mkat/L
4 -	mmol/L
5 -	mmol/L
6 -	mg/L
7 -	nkat/L
8 -	g/dL
9 -	mg/dL
10 -	IU/mL
11 -	mg/L
12 -	%

COMMUNICATIONS

CONFIG. RS-232:

- Bauds:	9600
- Timeout:	0
- Terminal number:	0

Appendix III. - Preventive maintenance plan

To maintain this semi-automatic analyzer in perfect conditions of use, we advise having it serviced every year by the Technical Assistance Service.

In each review, the following operations should be performed, in the order in which they are listed:

III.1. - Clean

- General cleaning of interior
- Flow cuvette

III.2. - Change

- Aspiration tube
- Dosing tube
- Interior waste output tube
- Exterior waste output tube

III.3. - Check

- Fan
- Keyboard
- Beep
- Screen
- Printer
- Serial channel
- Peristaltic pump motor
- Light intensity
- Darkness
- Readings stability

III.4. - Check

- Photometric accuracy
- Aspiration system adjustment
- Temperature adjustment

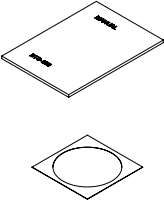
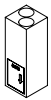
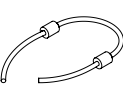
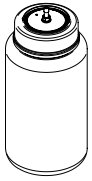
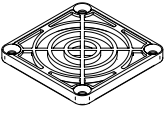
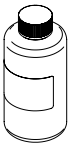
Appendix IV. - Password

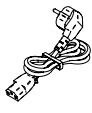
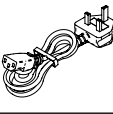
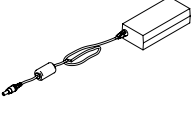
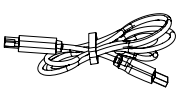
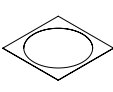
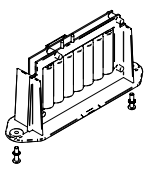
The PASSWORD to enter in the semi-automatic analyzer is:

251209

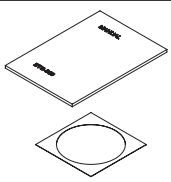
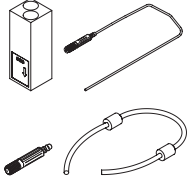
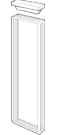
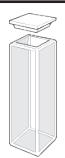
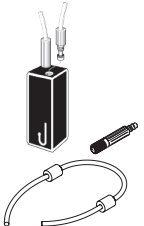
Appendix V. - Spares and accessories

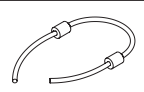
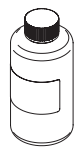
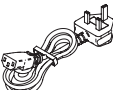
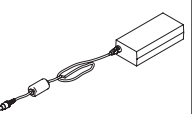
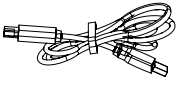
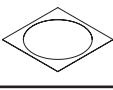
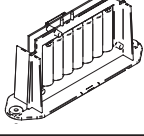
V.1. - List of accessories of BTS-350

Code	Description	
AC15416	Installation Manual (multilanguage) + User manual (CD-ROM)	
AC15161	Roll of paper	
AC15056	Flow cuvette	
AC15160	Teflon tube	
AC15057	Cuvette outlet fitting	
AC10413	Dosing tube of peristaltic pump	
AC10414	3x6 mm (200 cm) silicone tube	
AC10453	Test tube adapter	
AC10454	Waste bottle	
AC15111	Fan filter	
AC10415	Bottle of washing solution (100mL)	

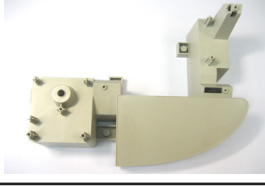
Code	Description	
CA10455	European mains cable	
CA10456	American mains cable	
AC14694	British mains cable	
AC15055	Voltage adapter	
FI14226	USB cable	
AC10243	"Semi-automatic analyzer PC" Software-CD	
AC14773	Battery pack	

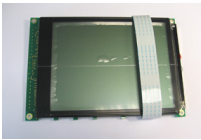
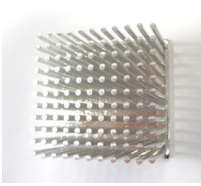
V.2. - List of accessories of Y-350

Code	Description	
AC17211	Installation manual (multi-language) + User manual (in CD-ROM)	
AC15161	Roll of paper	
AC17207	SOG glass flow cuvette (not used for the 280 nm wavelength)	
AC17208	SOG 1 mm glass cuvette	
AC17214	Spacer for 1 mm cuvette	
AC17209	10 mm quartz macro cuvette	
AC17210	10 mm quartz flow cuvette unit.	
AC17215	1 mm quartz flow cuvette unit.	
AC15160	Teflon tube	
AC15057	Cuvette outlet fitting	

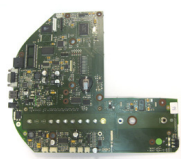
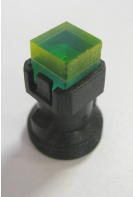
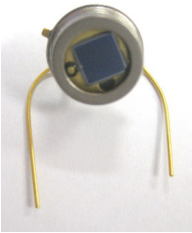
Code	Description	
AC10413	Dosing tube of peristaltic pump	
AC10414	3x6 mm (200 cm) silicone tube	
AC10453	Test tube adapter	
AC10454	Waste bottle	
AC15111	Fan filter	
AC10415	Bottle of washing solution (100mL)	
CA10455	European mains cable	
CA10456	American mains cable	
AC14694	British mains cable	
AC15055	Voltage adapter	
FI14226	USB cable	
AC10243	"Photometer PC" Software-CD	
AC14773	Battery pack	

V.3. - List of common spares parts for BTS-350 and Y-350

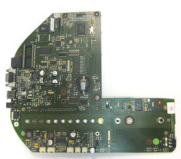
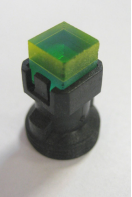
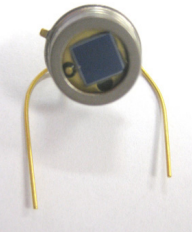
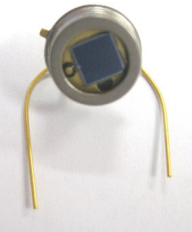
Code	Description	
CA15100	Top housing	
CA15101	Lower housing	
AC15102	Tray	
ZO15103	Printer support	
CA15104	Cuvette lid	
AC15105	Light guide	
AC15106	Guide tube	
TE15109	Keyboard support	
AC15112	Rubber legs	
AC15113	Waste output fitting	

Code	Description	
PC15114	Display	
TE15115	Keyboard	
IM15116	Printer	
ME15117	Peristaltic pump	
MO15120	Fan	
CE15121	Peltier	
AC15122	Temperature sensor	
AC15123	Radiator	
AC15094	tool for temperature with cuvette	

V.4. - Spare part for BTS-350

Code	Description	
PC15118	Electronic card BTS-350	
ME15119	Optical bench	
FI15128	340 Filter	
FI15129	405 Filter	
FI15130	505 Filter	
FI15131	535 Filter	
FI15132	560 Filter	
FI15133	600 Filter	
FI15134	635 Filter	
FI15135	670 Filter	
AC15136	340 LED	
AC15137	505 LED	
AC15138	405 LED	
AC15139	535 LED	
AC15140	560 LED	
AC15141	600 LED	
AC15142	635 LED	
AC15143	670 LED	
DI15145	Photodiode	

V.5. - Spare part for Y-350

Code	Description	
AC17243	Electronic card Y-350	
AC17244	Optical bench Y-350	
AC17245	280 Filter	
FI15128	340 Filter	
FI15129	405 Filter	
AC17246	420 Filter	
FI15130	505 Filter	
AC17247	520 Filter	
FI15132	560 Filter	
AC17248	620 Filter	
FI15134	635 Filter	
FI15135	670 Filter	
AC17249	750 Filter	
AC17250	280 LED	
AC15136	340 LED	
AC15138	405 LED	
AC15138	420 LED	
AC15137	505 LED	
AC15137	520 LED	
AC15140	560 LED	
AC15141	600 LED	
AC17251	620 LED	
AC15142	635 LED	
AC15143	670 LED	
AC17253	750 LED	
AC17252	Photodiode	

Appendix VI. - Changes Firmware and board versions for BTS-350

Relationship of changes in the firmware versions

Date	Version	Change
17/06/2009	1.02	First version of Firmware
18/03/2010	1.10	Added the following languages: French, Italian, Portuguese, Romanian, Polish, Serbian, Czech, Bulgarian, Russian, Chinese and Korean. Ability to update the firmware and the test parameters from an external USB memory. Included the reagent consumption choice on the screen of history. Ability to print and save data in an external USB memory at once.
05/10/2010	1.20	Correction of translation of some languages
14/04/2011	1.30	Add the arabic language
25/7/2012	1.31	Improve the management of peltier consumption.
30/10/2012	1.40	Add the Indonesian language. Add the option to activate/deactivate the concentration storage.
9/06/2014	1.42	Make compatible the change in the electronic board CIIM00037-08
30/4/2015	2.40	Add the new parameter dilution of sample for every calculation method Add the Ax+B comparison parameters. Add the blank to the calculation method kinetics and fixed time. Add for every calculation method the multipoint calibration.
19/12/16	2.41	Little internal modifications.

Relationship of changes in electronic board

Date	Version	Change
17/06/2009	CIIM00037-07	First version of board
10/06/2014	CIIM00037-08	Change the flash chip model for obsolete. Change the RS-232 connector to a USB type B. Improve the position of some components. Remove the internal metallic paint of the cover
10/10/2014	CIIM00037-09	Improve the internal position of some components. Ability to print and save data in an external USB memory at once.

Appendix VII. - Changes Firmware and board versions for Y-350

Relationship of changes in the firmware versions

Date	Version	Change
10/06/2014	2.00	First version of Firmware Y-350

Relationship of changes in electronic board

Date	Version	Change
10/06/2014	CIIM00037-08	First version of board
10/10/2014	CIIM00037-09	Improve the internal position of some components. Ability to print and save data in an external USB memory at once.

Appendix VIII. - Technical specifications



The manufacturer declines all responsibility for damages caused by incorrect use of the instrument.

Optical system	
Nominal range	-0.2 to 3.5 A
N° of wavelengths	8
Wavelengths installed (BTS-350)	340, 405, 505, 535, 560, 600, 635, 670
Wavelengths installed (Y-350)	280, 340, 405, 420, 505, 520, 560, 620, 635, 670, 750
Free wavelength positions (BTS-350)	2
Band width of each wavelength	10 ± 2 nm
Wavelength error ⁽¹⁾	± 2 nm
Digital resolution	0.0001 A for 2.0 A
Screen resolution	0.001 A
Baseline stability:	better than 0.001A in 60 minutes
Accuracy of the measurement ⁽¹⁾	CV $\leq 1\%$ at 0.1 A
	CV $\leq 0.1\%$ at 2 A
Trueness of the measurement ⁽¹⁾	$\pm 5\%$ at 0.1 A
	$\pm 2\%$ at 1.0 A
	$\pm 2\%$ at 2.0 A
	$\pm 5\%$ at 3.0 A
⁽¹⁾ for 340, 405, 505	

Thermostatic system	
Thermostatisation range	25° C to 40° C
Accuracy of the temperature	$\pm 0.5^\circ\text{C}$
Temperature stability	$\pm 0.2^\circ\text{C}$ in 30 minutes

Aspiration system	
Suction tube (external length)	approx. 100 mm
Flow cuvette	
Length of light passage	10 mm
Diameter of light passage	1.5 mm
Volume	18 μL
Quality of the glass	optical
Normal cuvettes	Macro, semimicro, micro
Test tubes	Round base with a diameter of 12 mm x 75 mm in length

Peristaltic pump	
Type of operation	Stepper motor
Nominal flow	10 mL/min
Aspiration volume	100 µL - 5000 µL
Cross contamination	< 3% at 200 µL
	< 1% at 300 µL
	< 0.5% at 400 µL
Waste bottle capacity	1L

Power supply system	
External power supply	
Voltage	100 V to 240 V AC
Frequency	50/60 Hz
Output voltage	15V CC
Output power	30W
Instrument power consumption	
Maximum	12 W without battery module
	24 W with battery module installed and in full load process
Average	5 W performing measurements
Minimum	2 W on standby
Optional battery system	
Capacity	2000 mAh
Minimum duration	2 hour with batteries fully charged

Dimensions and weight	
Maximum dimensions	420 mm x 350 mm x 216 mm (16.5 in x 13.7 in x 8.5 in)
Weight with battery pack	4290 gr. (9.43 lb)
Weight without battery pack	4000 gr. (8.8 lb)

Environmental conditions	
Use	indoor
Height	< 2000 m
Room temperature	10° C – 35° C
Maximum relative humidity	85%
Installation category (current surge category)	II
Pollution grade	2

Compliance with applicable directives and legislation BTS-350	
EC Directive 98/79 on medical devices for vitro diagnostics.	
Compliance with applicable directives and legislation Y-350	
EMC Directive: 2004/108 / EC	
Electrical Safety Directive: 2006/95 / EC	



- Certified Management System
- EN ISO 9001
- EN ISO 13485

Manufactured by: BioSystems, S.A.

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